



DEALER BEST PRACTICE SERIES

Managing Exchange
Component Inventory
Reduces Costs

Application Maintenance Site Management **Component Rebuild** Safety MARC Management

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1.0 Introduction

Most Mining customers require an inventory of exchange major components to support their production machines, primarily haul trucks and shovels. The inventory of exchange components can be owned by the customer, common for small fleets of equipment, or owned by the dealer as part of a major component exchange program, a common practice for large fleets of equipment.

While the purpose of a component exchange inventory is to minimize machine downtime, the costs to carry quantities of spare components can be significant. A typical set of components to support a fleet of Off Highway Trucks would include the engine, torque converter, transmission, differential, front and rear wheel groups, front and rear suspension cylinders, and hoist cylinders. For a 793, the seed cost of a set of new components for an exchange program exceeds \$900,000.

Determining how many exchange components are required to support a fleet of machines can depend on several key variables, but an accurate assessment considering all the factors impacting component demand can help to minimize and manage inventory costs.

This Best Practice summarizes factors to consider in the evaluation of component exchange inventory levels. Due to the complexity of factors to consider and the costs involved, component exchange inventory management makes a good 6Sigma project.

2.0 Best Practice Description

Many dealers supporting large mining customers use a “rule of thumb” guide to determine how many major components to stock in an exchange program based on the number of machines to be supported. The “rule of thumb” guidelines vary from one dealer to the next, but the most common practice is to stock one set of major components for every 10 machines. A few dealers follow a practice up to one set of components for every 15 - 18 machines.

While a “rule of thumb” guideline maybe easy to manage, there is a high cost to stock exchange components. Few dealers have actually quantified the “ideal” number of components “required” to meet expected consumption and provide backup for unplanned component repairs.

Most dealers think in terms of a “set” of components for machine, and not in terms of individual components. For example, a fleet may not consume as many fronts wheel groups as rear wheel groups due to the life of the components, therefore the same quantities of each component may not be required. Another



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example is that it may take a CRC twenty five days to rebuild an engine and only five days to rebuild a transmission. Perhaps an exchange inventory would not need as many transmissions as engines due to the shorter turnaround time from the CRC.

The above concepts could impact an exchange inventory management plan by adjusting the quantities stocked based on component life and CRC turnaround time. An inventory plan could look like the following:

Engine	One in stock for every 10 machines
Transmissions	One in stock for every 15 machines
Rear Wheel Group	Two (set) in stock for every 12 machines
Front Wheel Group	Two in stock for every 18 machines
Rear Suspensions	Two in stock for every 15 machines
Front Suspensions	Two in stock for every 20 machines

Evaluating requirements and adjusting quantities in inventory based on the requirement and key driving factors can save substantial investment costs, better utilize assets, all the while meeting the availability needs of the customer.



Managing an exchange component inventory becomes fairly simple once a process is established which determines quantities of components and which components are required to support a mining fleet of a given size. Establishing a component exchange management plan is best performed as a 6Sigma project due to the multiple variables, cost, and risk involved. Important to the process is establishing a process owner, someone responsible for the management of component exchange inventory.

Key factors to consider:

- **Customer Availability Requirements and Guarantees**

Guarantees with financial penalties involved pose a risk that may justify higher exchange inventory investment.

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- **Machine Population**

It is important to define how many machines of a given model and arrangement are being supported by major component exchange components. As populations change, stock quantities may need to be adjusted.

- **Component Life**

What is the local experience as to average life of each component? Components with shorter average component life may require more spare components than components with a long component life.

Customer fluid cleanliness and maintenance practices, along with application severity, have significant impact on component life.

- **Equipment Monitoring and Maintenance Practices**

A big factor is “planned” rebuilds versus “unplanned” rebuilds. Planned rebuilds can be scheduled with some flexibility and are easier to manage. Unplanned rebuilds are difficult to manage. In general, if less than 80% of the component rebuild activity is planned, there will be a requirement to increase inventories to protect against surprises.

Customer monitoring practices such as machine inspections, VIMS monitoring, particle count and SOS monitoring, and magnetic plug inspections impact the ability to plan before-failure rebuilds.

- **Component Reliability**

Components that have a history of high reliability pose less risk of an unplanned breakdown and therefore may require less protective inventory.

- **CRC Component Rebuild Turnaround Time**

This is the amount of time it takes a CRC to rebuild a component. For smaller components, which may only require 4 to 6 days to rebuild, the risk of machine downtime is lower and fewer components would be required for stock. If a spare exchange component is consumed, it can be replaced within a few days by the CRC.

Components with a long rebuild time, such as engines, carry increased risk and may dictate a higher stock level.

Shop turnaround time is defined as the number of days it takes to complete the rebuild of a component. This measurement takes all shop delays into account, such as ordering and receiving parts, single versus multiple shifts, machining /salvage delays, etc.

Note: Delays in the CRC should be reviewed for opportunities to improve the turnaround time. If engine block salvage is a major rebuild delay, it may be more economical to stock a spare block rather than a spare engine. The same may be true for other key parts, such as spindles.

- **Timing of Planned Rebuilds**

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When a new fleet of machines is sold, the initial stock of spare components is primarily needed to protect machine availability against the risk of an unplanned component breakdown. This “protective” stock can be a minimal stock.

The machines will not enter a planned component rebuild cycle until approximately two years after initial machine delivery. During a rebuild cycle, more exchange component stock may be required depending on how many components will be rebuilt, over what time frame, and how long it takes to rebuild and turnaround a component in the CRC.

- **Cat Reman**

If Caterpillar Remanufactured components are considered as a back up to CRC peak component rebuild cycles or unplanned rebuilds, then the time to order and receive a Reman component is a factor. Short lead-time can reduce inventory requirements by covering any temporary shortfalls. Long lead time would mean Reman components could be a back up plan for peak CRC workloads, but not for unplanned rebuilds.

- **Advance Scheduling**

The ability of a customer and dealer team to communicate and schedule rebuilds in advance can be an important critical success factor. This enables inventory assets to be assigned to a scheduled activity, and provides an advance warning if there is a coming shortage situation.

- **CRC Capacity and Volume**

Based on machine population and scheduled component rebuild activity, what is the normal number of any given component that would be in process at the same time? Is exchange component inventory level matched to a monthly in process volume?

- **Days at Risk**

How often does the inventory of a particular exchange component fall to zero in stock and how many days is the inventory at zero? This represents “days at risk” when if there is an unplanned breakdown, no spare component is available. Measuring the days at risk and establishing an “acceptable” level makes for an important part of the analysis.

Analyzing the above factors, an optimum inventory level can be determined which both supports the planned demand for exchange components and protects against the risk of unplanned breakdowns.

The following is a very simplified analysis meant to illustrate the basic principles.

A large mining customer orders a fleet of twenty five (25) 785C trucks. These are the first 785s in the territory and the dealer has no exchange components in stock. The machines will operate 7000 hours per year. The application is moderate. The customer uses the dealer for all component rebuilds and expects an exchange component program to support his fleet.

The machines will be delivered over a 12-month period, therefore the planned component rebuilds would be expected to occur over a 12-month period.

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The dealer needs to plan for an exchange component inventory, initially to protect against unplanned breakdowns and then to support the planned component rebuilds.

The factors impacting the analysis are contained in the following table.

Exchange Component Inventory - 25 X 785Cs										
Component	PCR Hours	Planned / Unplanned Percentage	Expected Unplanned Rebuilds	Expected Planned Rebuilds	Component Rebuild Cycle	Reman Shipping Time	Reman an Option	CRC Rebuild Time	Required Protective Inventory	Additional Stock for Rebuild Cycle
Engine	18,000 hrs	80 / 20	5	20	30 months	15 Days	No	25 Days	1	2
Transmission	12,000 hrs	70 / 30	8	17	20 months	15 days	No	8 Days	1	1
Wheel Gp - Rear	16,000 hrs	70 / 30	8	34	28 months	15 days	No	8 Days	1	2
Wheel Gp - Front	20,000 hrs	80 / 20	5	40	34 months	15 days	No	5 Days	1	1
Differential	16,000 hrs	80 / 20	5	20	28 months	N/A	N/A	5 Days	1	1
Torque Converter	18,000 hrs	80 / 20	5	20	30 months	15 Days	No	8 Days	1	1
Rear Suspension	14,000 hrs	80 / 20	5	40	24 months	5 Days	Yes	5 Days	1	1
Front Suspension	16,000 hrs	80 / 20	5	40	28 months	5 Days	Yes	5 Days	1	1
Hoist Cylinder	20,000 hrs	80 / 20	5	40	34 months	10 Days	No	8 Days	1	1

The stock to support the rebuild cycle was determined by the number of components to be rebuilt over the 12-month rebuild cycle and the CRC capacity and turnaround time. During the rebuild cycle, the CRC must complete two engine rebuilds per month. It takes 25 days to rebuild an engine, therefore two engines must be in process at the same time and thus two exchange components are required.

Two transmissions must also be rebuilt each month, however, a rebuilt only takes 8 days, so each transmission can be rebuild one after another. Only one exchange component is required to meet this schedule.

Some components are rebuilt as a set, such as both final drives and the differential. This may impact the number of exchange components required.

The most important aspect of managing exchange component inventory is to have a defined ownership and a defined process. The owner is someone responsible for inventory management and the process is the analysis steps and approval procedure to adjust inventory levels.

3.0 Implementation Steps

Due to the complexity of analyzing exchange inventory stocking requirements and the costs associated with carrying the inventory, it is highly recommended to use 6Sigma to conduct the analysis and establish a process for keeping it current. It is important up front to establish an owner for managing the exchange component inventory.

The goal is to support customer planned and unplanned exchange component requirements at a minimum carrying cost and acceptable inventory turnover.

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6Sigma steps include:

Define

Create a team, set goals and objectives of the team, look at any existing inventory management process, and determine customer requirements.

Measure

Collect data on mining machine populations, annual volume of planned and unplanned rebuilds, CRC component rebuild turnaround time, inventory levels and turnover, rebuild schedules, CRC capacity, and the data supporting the factors addressed in the above “Description” section. Analyze the risk for both the customers and the dealership.

Analyze

Develop solutions that match planned and unplanned component demand to inventory and CRC capacity using data collected in “Measure”. Identify all factors impacting demand and supply.

Implement

Establish the process, determine optimum stocking inventory, and adjust inventory over time to the target levels.

Control

Ensure that new process meets customer needs. Monitor and adjust inventories as machine population and other factors change.

4.0 Benefits

- Protect Mining customer machine availability and production.
- Create preference and loyalty for dealer CRC support.
- Manage CRC workflow.
- Minimize component exchange inventory investment and carrying costs.
- Support flat rate repair options.
- Grow rebuild business and revenue.
- Utilize dealer CRC and investment assets efficiently.
- Minimizes cost impact of product design changes on inventory components
- Improves inventory turnover ratio.

5.0 Resources Required

Exchange Component Inventory Management makes an excellent 6Sigma project that is relatively quick to complete and has immediate results. A key step to creating the process is to determine ownership, someone who is responsible for managing dealer component inventory practices.

6.0 Supporting Attachments / References

None.

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7.0 Related Best Practices

Improving CRC Rebuild Turnaround Time

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8.0 Acknowledgements

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